
COMP 3059 – Capstone Project I

Software Requirements Analysis and Design Assignment

This assignment is an overview to gather the software needs with requirements analysis and help to proceed with the design.

The requirements analysis helps to break down functional and non-functional requirements to a basic design view to provide a clear system development process framework. It involves various entities, including business, stakeholders and technology requirements.

The design is the activity following requirements specification and before programming. Software design usually involves problem solving and planning a software solution.

To work on this assignment, you could use the references and a sample template given below. The sample template can be customised to suit the nature of your project.

Reference Reading/Example to work on this Sprint document:

www.cse.msu.edu/~chengb/RE-491/Papers/SRSEExample-webapp.doc

Source for this template:

www.tricity.wsu.edu/~mckinnon/cpts322/cpts322-srs-v1.doc

1.0 Introduction

The GlamUp! Online Beauty Services Application is a mobile and web-based platform designed to connect clients with local freelance beauty professionals such as nail artists, lash technicians, and makeup artists. The system aims to make booking beauty services more convenient, affordable, and flexible for users while helping professionals gain visibility and manage their business digitally.

This document provides an overview of the software requirements and design for the GlamUp! system. It describes the system's purpose, scope, main functionalities, and overall context. The goal is to establish a clear understanding between the development team and stakeholders of what the system will deliver before proceeding to implementation.

The document includes both functional and non-functional requirements, data and process models, and other design considerations that define how the application will support its users and business goals.

1.1 Purpose

The purpose of this document is to define the high-level software requirements for the GlamUp! Online Beauty Services Application.

It focuses on what the system will do rather than how it will be implemented.

This document serves as a guide for developers, designers, and stakeholders by outlining the system's main objectives, features, and constraints.

It ensures that all project participants share a clear understanding of the functional and non-functional requirements needed to design and develop a reliable, user-friendly, and scalable application.

1.2 Scope

GlamUp! allows users to discover, book, and pay for beauty services from verified professionals near them. It also gives freelancers a platform to manage bookings and payments efficiently. The system aims to simplify appointment scheduling and provide a trusted, modern alternative to traditional salon bookings.

Definitions:

- Client: End-user seeking beauty services.
- Professional: Verified artist or technician offering services.
- Administrator: System manager responsible for user verification and content moderation.

Overview:

The remaining sections explain the system’s functionality, high-level models, and design process that guide GlamUp!’s development.

2.0 System Overview**2.1 Project Perspective**

GlamUp! is a stand-alone system with integration to third-party APIs like Google Maps (for location services) and Debit\Credit (for payments). It acts as a bridge between clients and professionals, supporting mobile and web interfaces.

2.2 Product Functions:

- User registration and login
- Search and filter professionals by location or service
- Booking and rescheduling appointments
- Secure online payment
- Ratings and reviews
- Admin dashboard for verification and monitoring

2.3 User Characteristics:

- **Clients:** Tech-savvy individuals looking for convenient, on-demand beauty services.
- **Professionals:** Freelancers who want exposure, flexible work schedules, and digital payment handling.
- **Admins:** Internal users ensuring quality, safety, and dispute resolution.

2.4 Constraints:

- Must comply with local business and payment policies.
- Internet connection required.
- Payment gateway fees and external API dependencies.

3.0 Functional Requirements

This section describes specific features of the software project. If desired, some requirements may be specified in the use-case format and listed in the Use Cases Section.

3.1.1 Functional Requirement 01 – *Client Registration*

Introduction	This feature allows new clients to create an account to browse services, view professionals, and book appointments.
Input	• Full name • Email address • Password • Phone number
Processing	• Validate input fields (email format, password length). • Check if the email is already in use. • Save user details securely in the database.
Outputs	• <u>Success</u>: Confirmation message (“Account created successfully”). • <u>Failure</u>: Error message (“Email already registered” or “Invalid input”). • Redirects to the Login page upon success.

3.1.2 Functional Requirement 02 – *Beauty Professional Registration & Verification*

Introduction	This feature allows beauty professionals (nail artists, lash technicians, makeup artists) to register and create a professional account. Their profiles must be verified before becoming active on the platform.
Input	• Full name • Email address • Password • Phone number • Business name (optional) • Proof of qualification (e.g., certification upload) • ID verification document (e.g., government-issued ID)
Processing	• Validate input fields. • Check if email already exists. • Store professional details securely. • Send verification request to admin.

	Admin reviews and approves/rejects the professional profile.
Outputs	<u>Success</u>: Message (“Registration submitted for verification”). <u>Failure</u>: Error messages for missing or invalid details. - Once approved, professional profile becomes visible to clients.

3.1.3 Functional Requirement 03 – User Log In (via Email & Password)

Introduction	This feature allows existing users to securely log into their accounts.
Input	- Email address - Password
Processing	- Verify credentials against stored user data. - Check if the account exists and password matches. - Establish a login session
Outputs	<u>Success</u>: Redirect user to the Home screen. <u>Failure</u>: Display error message (“Invalid credentials” or “Account not found”).

3.1.4 Functional Requirement 04 – User Log In (via Google Account)

Introduction	This feature allows users to log in using their Google account credentials, providing a faster and more convenient login experience without manually entering their email and password.
Input	- User selects “Sign in with Google” button. - Google OAuth credentials (retrieved securely through Google’s authentication API).
Processing	- Redirect to Google’s authentication service (OAuth 2.0). - User grants permission for <i>GlamUp!</i> to access basic profile information (name, email, profile picture). - System verifies the token returned by Google’s authentication API. - If the user exists in the database, log them in; otherwise, create a new user record.
Outputs	<u>Success</u>: User is redirected to the Home screen with an active session. <u>Failure</u>: Display an error message (“Sign-in failed, please try again”).

3.1.5 Functional Requirement 05 – Search (Home Screen for Client User)

Introduction	This feature enables client users to find professionals and services by entering keywords or service names in the search bar after landing on the homepage post-login.
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Input	Specific search keyword entered by the user
Processing	- Compare input text with database entries (service names, professional profiles, and descriptions). - Rank results based on relevance and popularity.
Outputs	<u>Success</u>: Display list of professionals or services matching the entered keyword. <u>Success</u>: Show relevant details such as service name, price, and professional rating. <u>Failure</u>: Displays error message, such as, “<i>No results found. Try using different keywords.</i>”, when system cannot find professionals or services matching the user’s search term.

3.1.6 Functional Requirement 06 – *Filter & Sort (Home Screen for Client User)*

Introduction	This feature allows client users to narrow down search results by applying specific filters such as category, price range, and distance on the homepage post-login.
Input	User-selected filter options (category, price range, distance).
Processing	- Retrieve the current list of search results. - Apply chosen filters to remove non-relevant entries. - Optionally allow sorting (e.g., by “Lowest Price” or “Highest Rating”).
Outputs	<u>Success</u>: Displays updated, filtered list of professionals based on selected filters. <u>Failure</u>: Displays error message if no professionals match the selected criteria.

3.1.7 Functional Requirement 07 – *View Professional Profile*

Introduction	This feature allows a client user to view detailed information about a selected beauty professional (e.g., nail artist, lash technician, or makeup artist). The profile displays essential details such as their name, bio, services, catalogue with prices, portfolio gallery, availability, and user reviews. It also provides action buttons for booking appointments or viewing ratings and reviews.
Input	User selects a professional from the search results or home screen.
Processing	- The system retrieves the professional’s data from the backend database, including: - Personal details (name, profile photo, short bio, years of experience).

	• Service catalogue (service names, descriptions, and prices). • Availability schedule (time slots). • Portfolio images showcasing past work. • Ratings and reviews submitted by clients. 2. The profile screen dynamically displays this data in a structured layout: • <u>Top section</u>: Professional’s photo, name and rating summary • <u>Service catalogue section</u>: List of services with prices and durations. • <u>Portfolio section</u>: Scrollable image gallery of the professional’s previous work. • <u>Buttons</u>: ◦ “Book Appointment” → Navigates to Appointment Booking page. ◦ “View Reviews” → Opens Ratings & Reviews page. 3. When the user clicks “Book Appointment” button: • The system passes the professional’s ID and selected service to the Appointment Booking screen. 4. When the user clicks “View Reviews” button: • The system fetches the corresponding professional’s reviews from the database and displays them.
Outputs	• Successful data retrieval: The professional profile is displayed with all available sections (portfolio, services, reviews). • User can take further actions by clicking the “Book Appointment” or “View Reviews” buttons.

3.1.8 Functional Requirement 08 – *Appointment Booking*

Introduction	This feature enables users to schedule beauty service appointments with their chosen professional by selecting a service, date, and available time slot. The system validates user selections and stores booking details securely for both the client and professional.
Input	• Selected professional ID (passed from Profile screen). • Selected service (from a dropdown or service list). • Date (via a DatePicker). • Time slot (via clickable time slot buttons).
Processing	1. The system retrieves available time slots for the selected professional and service. 2. The user selects a service, preferred date, and time.

	3. The system validates: • A service is selected. • Date is not in the past. • Selected time slot is available. 4. Once validated: • The booking data (service, date, time, user ID, professional ID) is temporarily stored. • The booking is confirmed and saved in the database. • The system generates a unique booking ID. 5. Notification triggers: • Confirmation message displayed to the user. • Professional receives a notification of the new booking.
Outputs	• <u>Success</u>: Confirmation message and redirection to a booking confirmation screen. • <u>Failure</u>: Displays error messages when either service, date and/or time are not selected.

3.1.9 Functional Requirement 09 – *Booking Confirmation*

Introduction	This feature provides users with a summary of their confirmed appointment details after successfully booking. It enhances user assurance by allowing them to review their selections before proceeding or returning to the home screen.
Input	• Data passed from the Booking screen (service, date, time, professional name, booking ID).
Processing	1. Retrieve booking data from the intent or database. 2. Format and display booking information in a readable summary view. 3. Provide user options to: • Return to the Home screen. • View or modify the appointment (optional future enhancement). 4. Store confirmation details locally for offline viewing (if enabled).
Outputs	• <u>Success</u>: Displays confirmation message: “<i>Your appointment is confirmed!</i>” along with the fields - service name, professional name, appointment date & time, booking reference number; there would also be

	an option to navigate back to Home or Book Another Appointment. • <u>Failure</u>: Displays error message if data transfer fails.
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3.1.10 Functional Requirement 10 – *Ratings & Reviews*

Introduction	This feature enables clients to provide feedback on completed services by rating their experience and writing an optional review. The ratings improve service visibility, while reviews help other users make informed decisions.
Input	• Rating (1–5 stars). • Optional written review text. • Professional ID and Booking ID (linked to completed appointment).
Processing	1. The system validates the input: • Ensures a rating is provided. • Limits review text to an acceptable character count (e.g., 150 words). 2. Links the rating and review to the corresponding professional and appointment. 3. Stores the feedback in the database. 4. Calculates and updates the professional's average rating dynamically. 5. Confirms review submission to the user.
Outputs	• <u>Success</u>: Success message: <i>"Thank you! Your review has been submitted."</i> and updated professional profile reflecting the new average rating. • <u>Failure</u>: Displays error message if user doesn't provide a rating.

3.1.11 Functional Requirement 11 – *Payment & Transaction Processing*

Introduction	This feature enables clients to make secure payments for beauty services booked through the GlamUp platform. The system supports multiple payment methods, including credit/debit cards and digital wallets, ensuring compliance with international payment and data protection standards such as PCI DSS and GDPR.
Input	• Service details (name, price, professional ID, booking ID). • User-selected payment method (e.g., card, Google Pay, PayPal, or GlamUp wallet).

	Card information (card number, expiry date, CVV) or digital wallet authorization token.
Processing	Initiate Payment: - When the user confirms a booking, the system retrieves the service price and displays the total amount payable. Payment Method Selection: - The user selects their preferred payment option. - For card payments, secure input fields are presented (using encryption). Transaction Authorization: - The system connects to a third-party payment gateway (e.g., Stripe, PayPal API) for transaction authorization. - User payment details are encrypted and transmitted securely via SSL/TLS. Payment Verification: - The gateway verifies available funds and returns a success or failure response. - The system validates the response and updates the booking record with a “Paid” status if successful. Transaction Record Creation: - Transaction details (transaction ID, amount, payment method, timestamp) are logged in the database. Notifications: - Confirmation messages are sent to both client and professional. - A digital receipt is generated for the client.
Outputs	<u>Success:</u> - Message: <i>“Payment successful! Your appointment is confirmed.”</i> - Booking status updated to “Confirmed / Paid.” - Receipt emailed or displayed in the app. <u>Failure:</u>

	• Error message when payment fails • Booking remains “Pending Payment.”
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3.1.12 Functional Requirement 12 – *Beauty Professional Dashboard*

Introduction	This feature provides registered and verified beauty professionals with a personal dashboard to manage their services, availability, appointments, and reviews. It serves as the main control panel for their business activities within the GlamUp platform.
Input	• Professional ID (used for data retrieval). • Service details (name, description, price, duration). • Availability time slots (entered manually or through a calendar view). • Profile and portfolio updates (images, bio, and certifications).
Processing	1. Retrieve existing professional data from the database upon login. 2. Allow professionals to: • Add, edit, or delete services offered. • Update prices and durations. • Set available dates and time slots for booking. • Upload or remove portfolio photos. • View upcoming and past appointments. 3. System validates all inputs (e.g., valid prices, non-overlapping time slots). 4. Save updates to the professional’s record in the database. 5. Reflect updates immediately in the client-facing profile.
Outputs	• <u>Success</u>: Displays confirmation messages upon updates made and shows upcoming bookings, total earnings and client feedback. • <u>Failure</u>: Displays error message if data transfer fails or invalid input is given.

3.1.13 Functional Requirement 13 – *Beauty Professional Booking Management*

Introduction	This feature enables beauty professionals (nail artists, lash technicians, makeup artists) to manage all appointment-related activities within their GlamUp dashboard. Professionals can view new and upcoming bookings, update appointment statuses, reschedule sessions, or cancel bookings if
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	necessary. The feature ensures smooth coordination between clients and professionals.
Input	1. Professional login credentials (for authentication). 2. Booking details retrieved from the database (booking ID, client name, date, time and service type). 3. Professional actions such as: • Accepting or declining a new booking. • Updating the status of an appointment (e.g., <i>Pending</i> → <i>Confirmed</i> → <i>Completed</i>). • Editing time or rescheduling a booking. • Cancelling an appointment with reason.
Processing	1. The system retrieves all bookings associated with the logged-in professional from the database. 2. Professionals can filter bookings by: • Status: Pending, Confirmed, Completed, Cancelled. • Date: Today, Upcoming, Past. 3. The system allows the professional to: • View detailed booking information (client name, service, date/time, price). • Accept or decline new booking requests. • Update booking status after the appointment. • Modify appointment time if the client agrees (optional). 4. All updates trigger automatic notifications to the client (e.g., “Your appointment has been confirmed” or “Your booking has been rescheduled”). 5. The system logs all booking activities for recordkeeping and reporting.
Outputs	<u>Success:</u> • Updated booking list visible on the professional’s dashboard. • Real-time notifications sent to clients upon any change in booking status. • Confirmation messages displayed to professionals. <u>Failure:</u> Error messages when bookings fail to load or update.

3.1.14 Functional Requirement 14 – *Beauty Professional Payment Management*

Introduction	This feature enables beauty professionals to manage all payments received through the <i>GlamUp!</i> platform. It provides a digital and transparent payment system, removing the need for cash handling and ensuring that all transactions are securely processed and tracked.
Input	- Professional login credentials (authentication). - Booking and transaction data retrieved from the database. - Filters for viewing transactions (by date, client, or service).
Processing	- Retrieve all completed and pending payments for the logged-in professional. - Display payment details such as: - Client name and service booked. - Total amount paid or pending. - Payment date and method (card, wallet, etc.). - Transaction ID. - Automatically update payment status once a client successfully completes a transaction. - Allow professionals to track total earnings, pending deposits, and completed payouts. - Generate and display digital receipts for recordkeeping. - Provide an overview of earnings analytics (optional future enhancement).
Outputs	<u>Success:</u> - Display list of transactions categorized as <i>Pending</i>, <i>Completed</i>, or <i>Refunded</i>. - Summary statistics (e.g., total earnings this week/month). - Notifications upon successful payments. <u>Failure:</u> - Error message when system fails to retrieve the payments

3.1.15 Functional Requirement 15 – *Booking Deposit Management*

Introduction	This feature enables beauty professionals to require clients to pay a deposit when booking an appointment. The deposit secures the booking and discourages last-minute cancellations. It integrates seamlessly with the platform's payment system to automatically collect and record deposit
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	payments.
Input	• Service details (service name, total price, and duration). • Deposit percentage or flat amount set by the beauty professional. • Client-selected service and preferred date/time for booking. • Payment method for the deposit (card, wallet, or digital payment gateway).
Processing	1. Deposit Setup by Professional: • The beauty professional defines a deposit rule (e.g., 20% of total service cost or a fixed amount). • The system stores this rule in the professional's account profile. 2. Deposit Calculation: • When a client selects a service, the system automatically calculates the required deposit based on the set rule. 3. Payment Collection: • The client pays the deposit amount through the integrated payment gateway. • The payment gateway authorizes and records the transaction securely (PCI DSS-compliant). 4. Booking Confirmation: • The system confirms the appointment only after deposit payment is successful. • A confirmation message and receipt are sent to both the client and professional. 5. Cancellation or No-show Handling: • If a client cancels within the allowed time window, the system may refund the deposit according to the cancellation policy. • If the cancellation happens too close to the appointment time or the client does not show up, the deposit is forfeited to the professional.
Outputs	<u>Success:</u> • Confirmation message for client • Payment receipt showing deposit amount, transaction ID, and booking details • Updated professional earnings showing deposit credited • Notification sent to professional regarding successful deposit

	<u>Failure:</u> Error message when system fails to receive transaction due to network issue or payment being declined
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3.1.16 Functional Requirement 16 – *Private Messaging System*

Introduction	The Private Messaging feature allows clients and beauty professionals to communicate securely within the <i>GlamUp!</i> app. This feature supports sharing appointment details, special requests, or post-service feedback without revealing personal contact information, maintaining both privacy and professionalism.
Input	- Logged-in user (client or beauty professional). - Recipient user ID (linked to the chat thread). - Message text input (and optional attachments such as images).
Processing	1. Message Creation: - The user composes a message and presses <i>Send</i>. - The system timestamps the message, associates it with the conversation thread, and stores it in the backend database. 2. Message Delivery: - The system pushes the message to the recipient's inbox in real time using real-time communication service. - Notifications are generated for new incoming messages. 3. Conversation Management: - Each conversation is tied to either an active booking or a prior interaction between the same client and professional. - Users can view their chat history and search within messages. - Unread messages are visually marked until opened. 4. Data Protection: - No personal contact info (phone numbers, emails) is shared between users. - Messages are encrypted in transit and stored securely in the database.
Outputs	<u>Success:</u> - Display of message threads between client and professional. - Real-time updates (new messages appear instantly without

	refreshing). • Notifications for incoming messages. <u>Failure:</u> • Error messages when messages aren't sent due to connectivity issues, chat service is unavailable and images aren't uploaded to the conversation.
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3.1.17 Functional Requirement 17 – *Reporting User*

Introduction	This feature allows both clients and beauty professionals to report inappropriate behavior, harassment, fraud, or service issues directly through the <i>GlamUp!</i> platform. It ensures a safe and respectful environment by enabling users to flag misconduct for administrative review. Clients can report professionals for: • Unprofessional conduct, harassment, or inappropriate communication. • Poor service quality or fraudulent profiles. • Issues with overcharging or unsafe products. Professionals can report clients for: • Repeated no-shows or last-minute cancellations. • Harassment or offensive communication. • Fraudulent payment behavior.
Input	• Reporter's user ID (client or professional) • Reported user ID (client or professional) • Type of report • Description of the incident via text field input • Optional attachments
Processing	Accessing Report Option: • Users can access the "<i>Report User</i>" button from a profile page, chat screen, or booking history. Report Form Submission: • The system displays a form for the reporter to select a reason and provide details. • Upon submission, the report is validated for required fields and sent to the admin backend.

	3. Backend Handling: - The system logs the report with a timestamp, reporter and reported user IDs, and report category. - A confirmation is sent to the reporter. - The admin dashboard receives the report for manual review. 4. Admin Review Process: - The admin can view the report details, review associated chat logs or booking records, and decide on actions (e.g., issue warning, suspend user, or dismiss report).
Outputs	<u>Success:</u> - Confirmation message to reporter - Notification to admin dashboard <u>Failure:</u> - Error messages when user tries to submit the form with incomplete mandatory fields or tries to attach unsupported files.

3.1.18 Functional Requirement 18 – *Admin Analytics Dashboard*

Introduction	The Analytics Dashboard provides admins with a visual overview of platform metrics, including user growth, booking frequency, revenue, and activity distribution.
Input	System analytics data (user count, booking records, ratings, payments).
Processing	- Aggregate data from multiple modules (users, bookings, payments). - Generate charts for metrics such as daily active users, top-rated professionals, and total revenue. - Refresh dashboard periodically (e.g. every 24 hours).
Outputs	<u>Success:</u> - Interactive graphs and summaries. - Exportable reports (PDF/CSV). <u>Failure:</u> - Error message due to failed analytics refreshing

3.1.19 Functional Requirement 19 – *Beauty Professional Account Verification*

Introduction	This feature allows admins to verify new beauty professional accounts to ensure that only legitimate and qualified artists are approved to operate on the platform.
Input	• New professional's registration data (name, business name, certifications, ID proof). • Uploaded verification documents.
Processing	• Admin reviews submitted documents. • The system cross-checks details for completeness and flags missing information. • Admin either approves or rejects the application. • The system updates the professional's account status (verified / pending / rejected).
Outputs	<u>Success:</u> • Notification to the professional upon finished processing of account verification. • Dashboard status updated for admin records. <u>Failure:</u> • Displays error message due to unsupported attachment type of validation documents or network issues.

3.1.20 Functional Requirement 20 – *Admin Report Management*

Introduction	This functionality allows admins to monitor and act upon user reports submitted against clients or professionals to maintain platform safety and trust.
Input	• Report details (reporter ID, reported user ID, description, evidence).
Processing	• Retrieve reports from the database. • Display report details with user information and booking history. • Admin reviews, categorizes, and updates report status (resolved / under review / dismissed). • Admin can take actions like <i>warning</i>, <i>temporary suspension</i>, or <i>account removal</i>.
Outputs	<u>Success:</u>

	• Confirmation message sent to admin after report is resolved • Notification to both users (if applicable) about the outcome <u>Failure:</u> • Report details are not loaded due to network issues • Unable to save action due to network issues
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3.1.21 Functional Requirement 21 – *Admin Service Category Management*

Introduction	The Admin Service Category Management feature enables administrators to create, modify, organize, and remove service categories within the <i>GlamUp!</i> platform in order to keep the service listings up to date, relevant, and easy to navigate for both clients and beauty professionals.
Input	1. Admin login credentials (authentication required). 2. Category details including: • Category name (e.g., “Nails”, “Makeup”). • Description of the category. • Sub-services (e.g., “Gel Manicure”, “Bridal Makeup”, “Lash Lift”).
Processing	• Create Category: 1. Admin enters category information through a dashboard form. 2. The system validates that the name is unique and fields are complete. 3. Category is stored in the database and becomes visible to users upon approval. • Update Category: 1. Admin selects an existing category and edits details such as name, description, or sub-services. 2. The system updates the database and reflects changes across client and professional views in real time. • Manage Sub-services: 1. Admin can create or modify sub-services under each category (e.g., “Acrylic Nails” under <i>Nails</i>). • Sync with Front-End: 1. Changes in the service categories are automatically synced with the client and professional interfaces. 2. Updated categories become available in search filters and professional onboarding forms.
Outputs	<u>Success:</u> • Confirmation message upon successful category creation • Updated user interfaces for clients and beauty professionals

	<u>Failure:</u> • Failed database update • Error message due to duplicate category name or missing required fields
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3.1.22 Functional Requirement 22 – *Admin Dispute Management*

Introduction	The Admin Dispute Management feature enables administrators to handle conflicts between clients and beauty professionals efficiently. The feature provides tools to track, evaluate, and resolve disputes transparently.
Input	• Dispute ticket information (generated automatically from the booking or payment record). • Admin notes and resolution decision.
Processing	Ticket Generation: • A dispute ticket is automatically created when a user reports an issue related to a booking. • The system links the ticket to the associated booking and user accounts. Ticket Review: • Admin retrieves and reviews the dispute details. • The system displays a summary of the transaction, payment status, chat history, and any submitted evidence. Investigation & Communication: • Admin may contact both parties through an internal message thread for clarifications. • Admin logs all communication for future reference. Resolution Decision: • Based on policy guidelines, the admin can: ○ Approve a refund (partial or full). ○ Credit future booking vouchers. ○ Issue a warning to either party. ○ Close the case as “resolved” or “dismissed.” Update System Records: • The system updates the dispute status and sends resolution notifications to both users. • The payment and booking records are adjusted accordingly (e.g.,

	refund processed).
Outputs	<u>Success:</u> • Confirmation message when dispute resolved • Updated dispute ticket status <u>Failure:</u> • Error message due to failure to load dispute details

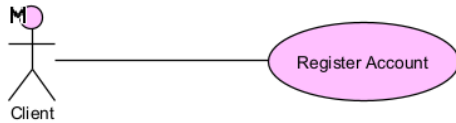
3.1.23 Functional Requirement 23 – *Admin Financial Reporting*

Introduction	This feature allows administrators to oversee and audit all financial activities within the <i>GlamUp!</i> platform. It ensures transparency and accuracy across transactions, including client payments, <i>professional</i> payouts, refunds, deposits, and commissions.
Input	• Transaction records (payments, deposits, refunds, and payouts). • Filters for category, professional, client, or transaction type. • Commission and tax rate data.
Processing	• Fetch and summarize transaction logs. • Display detailed revenue breakdown by category, date, and professional. • Highlight failed or pending transactions for investigation.
Outputs	<u>Success:</u> • Downloadable financial summary reports. • Dashboard indicators (e.g., “Total Revenue This Month”). <u>Failure:</u> • Error message due to failure to retrieve payment records.

3.2 Use Cases

3.2.1 Use Case 1: Client Registration

Diagram:



Brief Description:

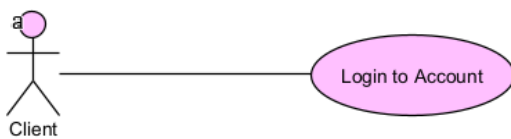
A new client creates an account by entering personal details such as name, email, and password, enabling access to *GlamUp!*'s booking and search features.

Initial Step-by-Step Description:

1. Client opens the *GlamUp!* app and selects "Sign Up."
2. System displays registration form.
3. Client enters required details (name, email, password).
4. System validates the input fields.
5. If valid, system saves the account and sends a confirmation email.
6. System confirms successful registration and redirects to the login screen.

3.2.2 Use Case 2: Client Login (including Google Sign-In)

Diagram:



Brief Description:

Clients log into *GlamUp!* using email/password credentials or their linked Google account for convenience and faster access.

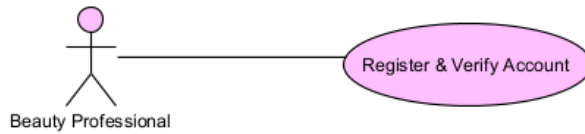
Initial Step-by-Step Description:

1. Client opens the GlamUp app and selects "Login."
2. System displays two options: Email/Password or Google Sign-In.
3. Client enters credentials or selects their Google account.
4. System authenticates credentials or verifies Google token.

5. If successful, the system loads the user's home dashboard.
6. If unsuccessful, an error message appears: "Invalid credentials."

3.2.3 Use Case 3: Professional Registration & Verification

Diagram:



Brief Description:

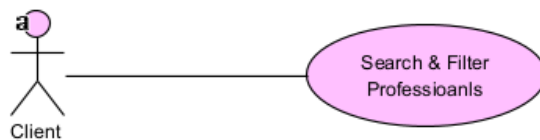
Beauty professionals register and upload verification documents (certifications, ID) to ensure authenticity before activating their profiles.

Initial Step-by-Step Description:

1. Professional selects "Register as Beauty Professional."
2. System displays a form requesting business details and certification uploads.
3. Professional completes the form and submits documents.
4. System saves the data and marks the account as "Pending Verification."
5. Admin reviews the application and either approves or rejects it.
6. System updates the professional's status and sends a notification email.

3.2.4 Use Case 4: Search & Filter Services

Diagram:



Brief Description:

Clients search for beauty professionals by service category, price range, or distance to find the most suitable provider.

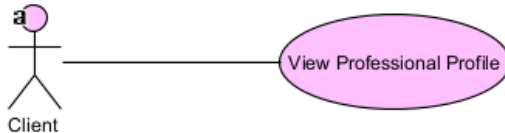
Initial Step-by-Step Description:

1. Client opens the search bar from the home screen.
2. System displays filter options: category, price, and distance.
3. Client enters keywords or applies filters.

4. System retrieves matching professionals from the database.
5. Results are displayed with ratings, prices, and availability.
6. Client selects a professional profile to view more details.

3.2.5 Use Case 5: View Professional Profile

Diagram:



Brief Description:

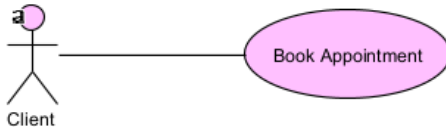
Clients view detailed professional profiles including portfolio, service catalog, price list, reviews, ratings, and a booking button.

Initial Step-by-Step Description:

1. Client selects a professional from the search results.
2. System displays the professional's profile page.
3. Profile includes photos, service list, and price details.
4. Client scrolls to view reviews and ratings.
5. Client clicks "Book Appointment" or "Send Message" as next steps.

3.2.6 Use Case 6: Book Appointment

Diagram:



Brief Description:

Clients schedule a beauty service by selecting a professional, choosing a service, date, and time slot, and confirming their booking.

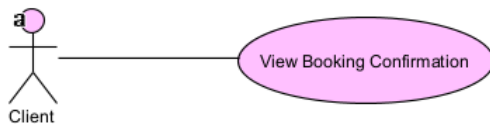
Initial Step-by-Step Description:

1. Client opens a professional's profile.

2. Client clicks the “Book Appointment” button.
3. System displays available services, dates, and time slots.
4. Client selects the preferred service, date, and time.
5. System validates the selections and checks availability.
6. Client confirms the booking.
7. System generates a booking confirmation and updates the professional’s schedule.

3.2.7 Use Case 7: Booking Confirmation Display

Diagram:



Brief Description:

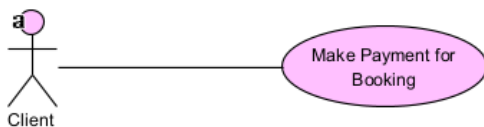
Bookings details are displayed to the client after a successful appointment booking, including service, professional name, date, and time.

Initial Step-by-Step Description:

1. System generates a confirmation message after booking completion.
2. System displays booking details (service, professional, date, time).
3. Client reviews the details.

3.2.8 Use Case 8: Payment Processing

Diagram:



Brief Description:

Clients make secure payments for their bookings through in-app payment gateways, ensuring compliance with PCI DSS and GDPR.

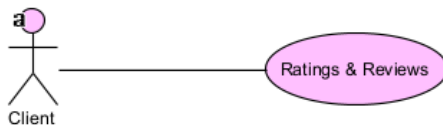
Initial Step-by-Step Description:

1. Client proceeds to the payment screen after booking.

2. System displays available payment options (credit/debit card, Google Pay, etc.).
3. Client enters payment details or selects a saved method.
4. System validates and encrypts the transaction.
5. Payment gateway processes the amount.
6. System confirms payment and updates the booking status to “Paid.”

3.2.9 Use Case 9: Ratings & Reviews

Diagram:



Brief Description:

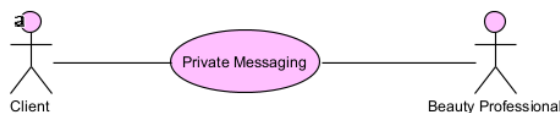
Clients rate and review professionals after their appointments to help others make informed decisions.

Initial Step-by-Step Description:

1. Client opens their past bookings list or professional profile.
2. System displays “Ratings” option.
3. Client selects a booking/profile and enters a rating (1–5 stars) and comment.
4. System validates input and submits the review.
5. The professional’s average rating is updated automatically.
6. Client receives a confirmation: “Thank you for your feedback!”

3.2.10 Use Case 10: Private Messaging

Diagram:



Brief Description:

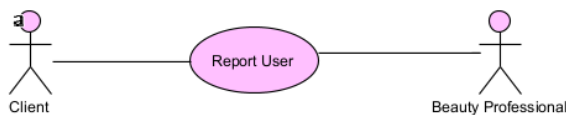
Clients and professionals communicate directly through in-app messaging to discuss booking details, customization, or service preferences.

Initial Step-by-Step Description:

1. Client selects a professional and clicks “Send Message.”
2. System opens a private chat thread between client and professional.
3. Users exchange text messages and attachments (e.g., reference photos).
4. System stores all messages securely in the database.
5. Users receive notifications for new messages.
6. Either party can close the chat after service completion.

3.2.11 Use Case 11: Report User

Diagram:



Brief Description:

Clients or beauty professionals report inappropriate behavior, fraudulent activity, or policy violations to the admin team for review.

Initial Step-by-Step Description:

1. User opens another user’s profile or chat window.
2. User clicks “Report User.”
3. System opens a report form requesting details and evidence (text or image).
4. User submits the report.
5. System saves the report and forwards it to the admin dashboard.
6. Admin reviews and takes appropriate action (warning, suspension, or ban).
7. User receives confirmation: *“Your report has been submitted for review.”*

3.2.12 Use Case 12: Professional Booking Management

Diagram:



Brief Description:

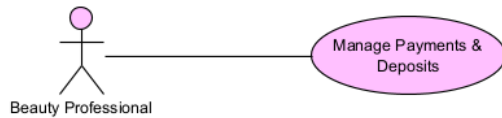
Beauty professionals view, accept, reschedule, or cancel client bookings directly from their dashboard.

Initial Step-by-Step Description:

1. Professional logs into their account and accesses the “Bookings” section.
2. System displays upcoming and pending appointment requests.
3. Professional reviews details (service, client name, time).
4. Professional accepts, declines, or proposes a new time.
5. System updates the booking status accordingly and notifies the client.
6. System synchronizes changes to both users’ schedules.

3.2.13 Use Case 13: Payment & Deposit Management

Diagram:



Brief Description:

Beauty professionals manage incoming payments and set deposit requirements to reduce no-shows and late cancellations.

Initial Step-by-Step Description:

1. Professional navigates to the “Payments” section in their dashboard.
2. System displays recent transactions and pending payouts.
3. Professional sets deposit percentage or flat rate for selected services.
4. System updates service details accordingly.
5. When a booking is made, the client pays the deposit upfront.
6. System tracks and credits the balance payment after service completion.

3.2.14 Use Case 14: Professional Profile and Portfolio Management

Diagram:

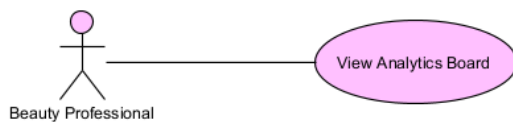


Brief Description:

Beauty professionals create and update their public profile with service details, pricing, availability, and a photo gallery to attract clients.

Initial Step-by-Step Description:

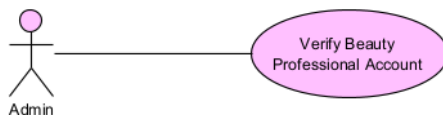
1. Professional logs in and accesses “My Profile.”
2. System displays editable fields: name, bio, pricing, service list, and images.
3. Professional uploads portfolio photos and updates details.
4. System validates the inputs and saves changes.
5. Updated information appears immediately on the client-facing profile.
6. System confirms: “*Profile updated successfully.*”

3.2.15 Use Case 15: Professional Analytics Board**Diagram:****Brief Description:**

Beauty professionals view statistics about their business performance — including total bookings, revenue, ratings, and most popular services — to support better decision-making.

Initial Step-by-Step Description:

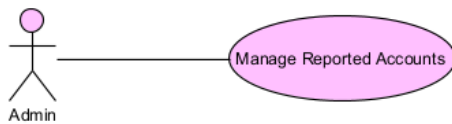
1. Professional logs in and opens the “Analytics” section.
2. System retrieves recent performance data from the database.
3. Dashboard displays charts for bookings, income, and average rating.
4. Professional filters data by week, month, or service category.
5. System updates charts dynamically based on filters.
6. Professional reviews performance trends and insights.

3.2.16 Use Case 16: Admin Beauty Professional Verification**Diagram:****Brief Description:**

Admins review and approve newly registered beauty professionals by verifying their documents and credentials before activating their accounts.

Initial Step-by-Step Description:

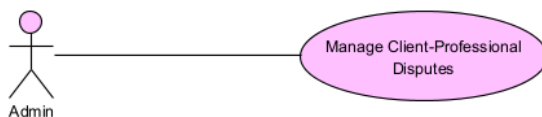
1. Admin logs into the *GlamUp!* Admin Dashboard.
2. System displays pending professional verification requests.
3. Admin reviews each applicant's documents and details.
4. Admin approves or rejects the application.
5. System updates the professional's account status (Verified / Rejected).
6. Notification is sent to the professional regarding the decision.

3.2.17 Use Case 17: Admin Report Management**Diagram:****Brief Description:**

Admins review user reports against clients or professionals and take necessary actions to maintain trust and safety on the platform.

Initial Step-by-Step Description:

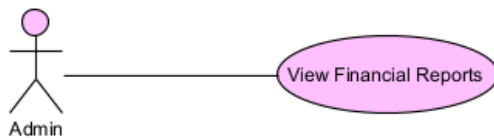
1. Admin accesses the “Reported Users” section.
2. System retrieves all active reports submitted by users.
3. Admin selects a report to review its details and evidence.
4. Admin decides to issue a warning, suspend, or ban the reported user.
5. System updates report status (Resolved / Dismissed / Action Taken).
6. Both reporter and reported users are notified of the outcome.

3.2.18 Use Case 18: Admin Dispute Management**Diagram:****Brief Description:**

Admins handle and resolve disputes between clients and beauty professionals related to bookings, cancellations, or payments.

Initial Step-by-Step Description:

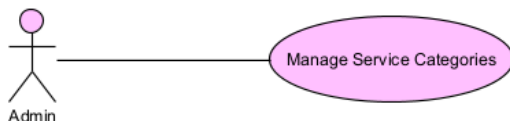
1. Admin opens the “Disputes” dashboard.
2. System lists all pending dispute tickets.
3. Admin reviews the case details, attached evidence, and chat history.
4. Admin communicates with both parties if clarification is needed.
5. Admin decides on resolution (refund, warning, dismissal).
6. System updates dispute status and sends notifications to both users.
7. Resolution record is stored for auditing and reporting.

3.2.19 Use Case 19: Admin Financial Reporting**Diagram:****Brief Description:**

Admins monitor financial transactions including client payments, refunds, and professional payouts, ensuring accuracy and compliance with financial standards.

Initial Step-by-Step Description:

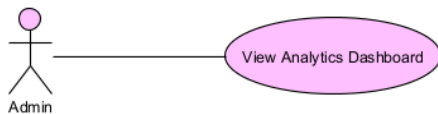
1. Admin navigates to the “Financial Reports” section.
2. System fetches transaction data from the payment gateway.
3. Dashboard displays revenue summaries, commissions, and payout data.
4. Admin filters data by category, professional, or date range.
5. System generates charts and downloadable reports (PDF/CSV).
6. Admin reviews for discrepancies or anomalies.
7. Admin exports data for auditing or taxation records.

3.2.20 Use Case 20: Admin Service Category Management**Diagram:****Brief Description:**

Admins create, edit, or remove service categories (e.g., Nails, Lashes, Makeup) to organize available offerings for clients and professionals.

Initial Step-by-Step Description:

1. Admin logs into the GlamUp Admin Dashboard.
2. Admin navigates to the “Service Categories” section.
3. System displays a list of all active categories.
4. Admin selects an option to add, edit, or delete a category.
5. Admin enters category name, description, and optional image.
6. System validates input and saves changes to the database.
7. Updated categories appear instantly for users in the app.
8. Admin receives confirmation: “*Category updated successfully.*”

3.2.21 Use Case 21: Admin Analytics Dashboard**Diagram:****Brief Description:**

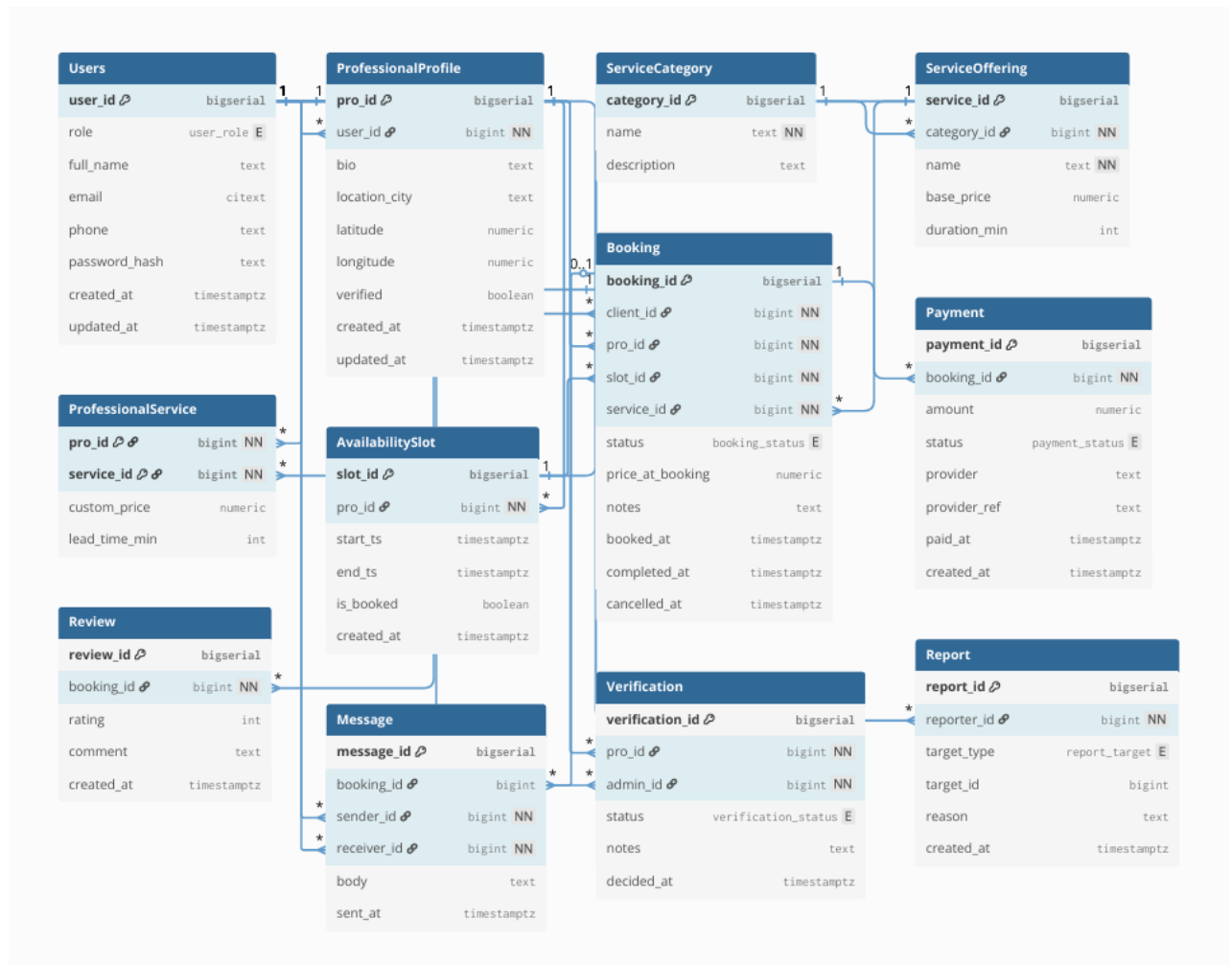
Admins can access a visual dashboard showing platform metrics such as user growth, booking activity, revenue, and category popularity to support decision-making.

Initial Step-by-Step Description:

1. Admin logs in and navigates to the “Analytics” section.
2. System aggregates data from multiple modules (users, bookings, and payments).
3. Dashboard displays charts for user signups, active bookings, and revenue trends.
4. Admin filters results by category, region, or time period.
5. System dynamically updates charts based on the selected filters.
6. Admin reviews insights for performance monitoring or reporting.
7. Optionally, Admin exports data as a PDF or CSV report.

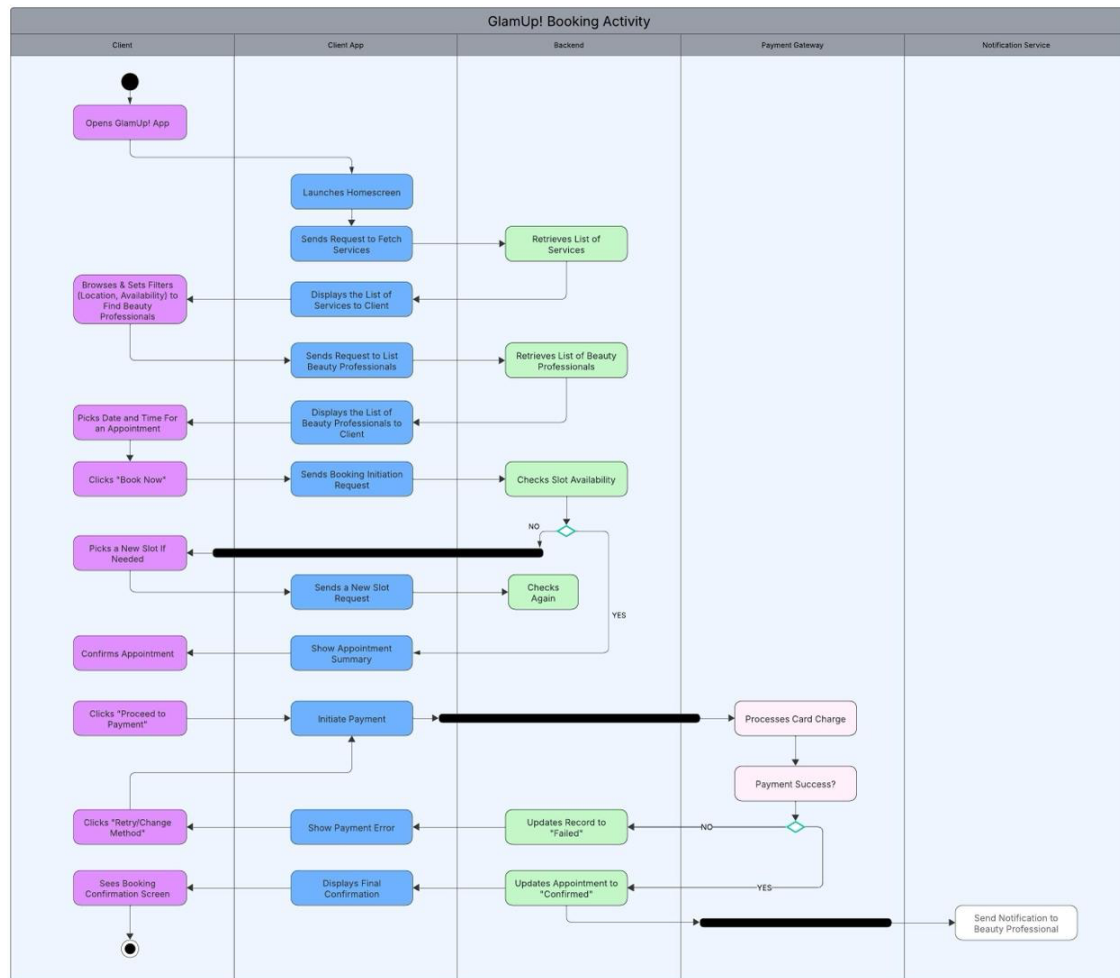
3.3 Data Modelling and Analysis

3.3.1 Normalized Data Model Diagram



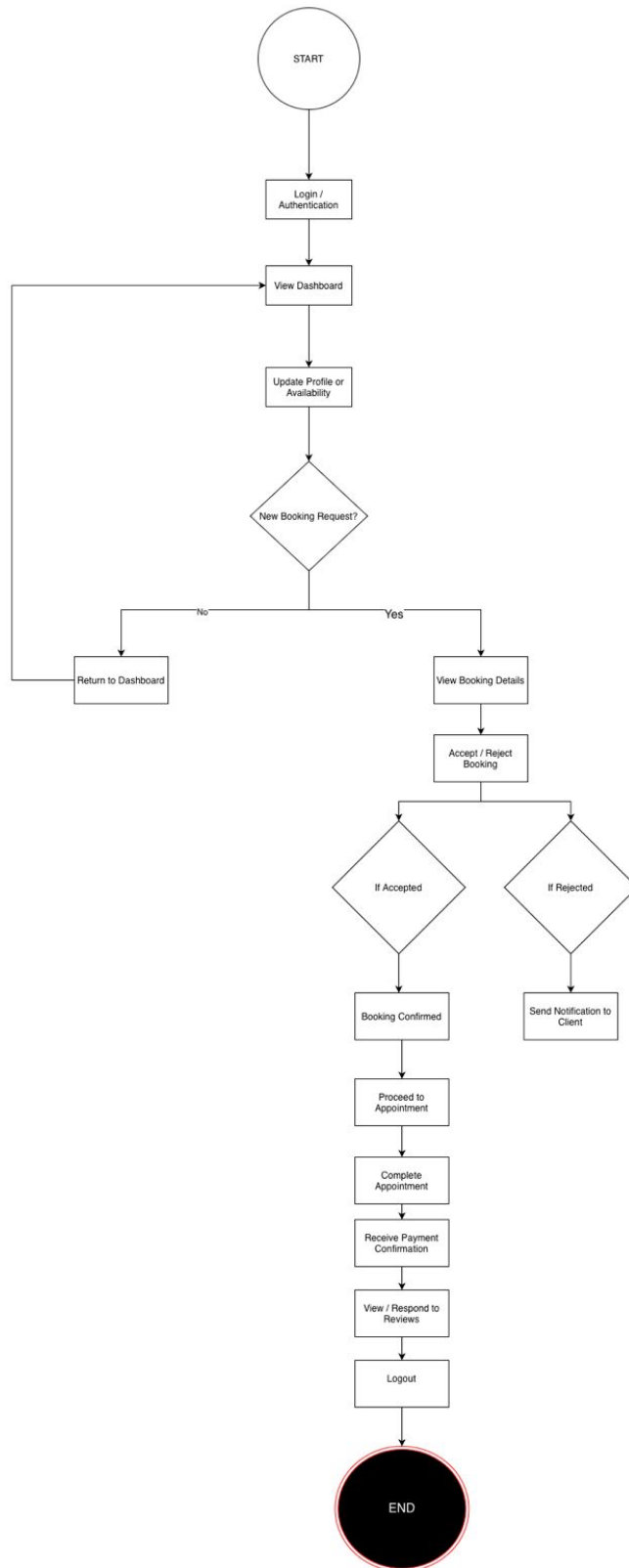
3.3.2 Activity Diagrams

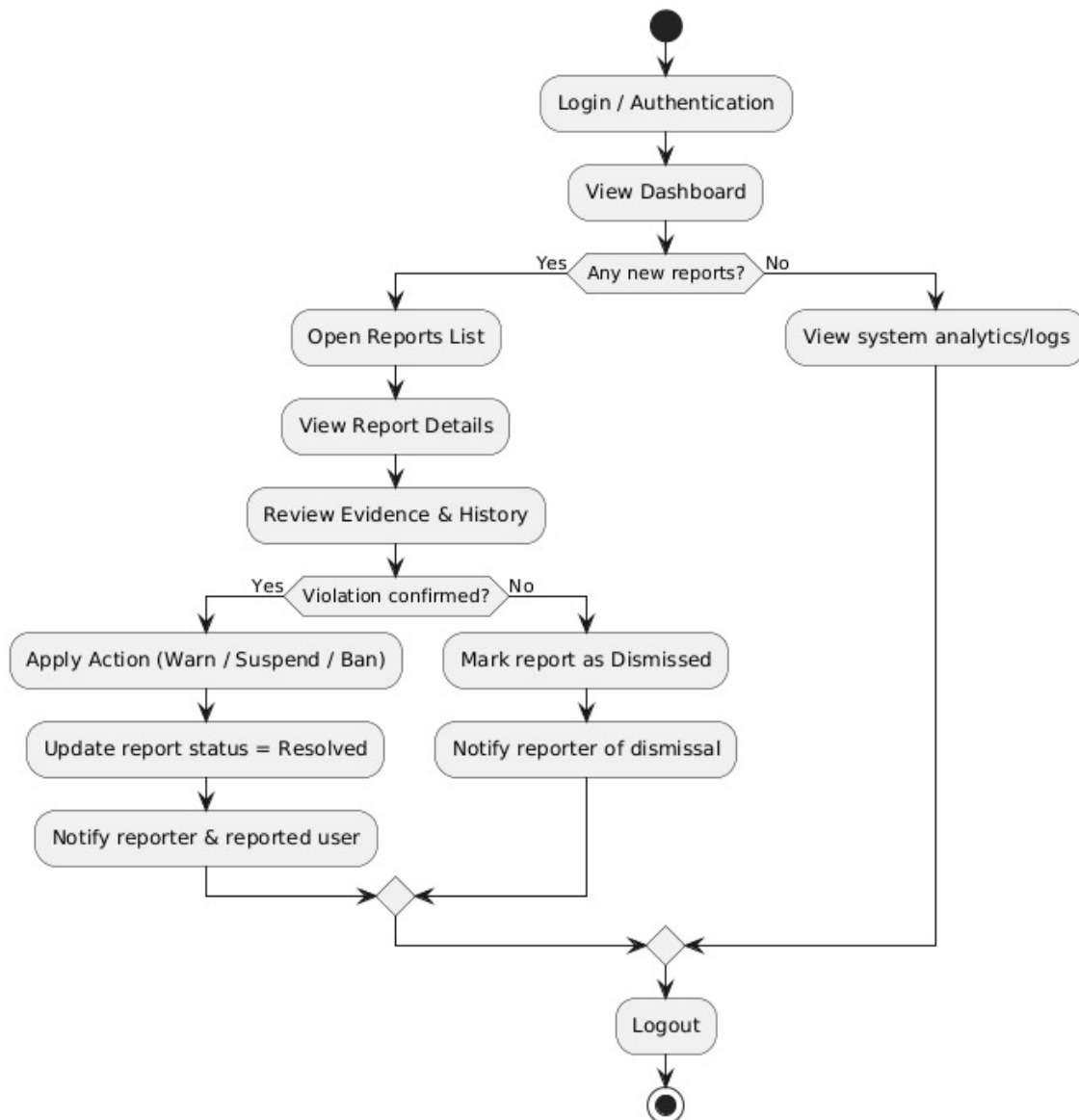
Booking Activity:



Beauty Professional Activity Diagram:

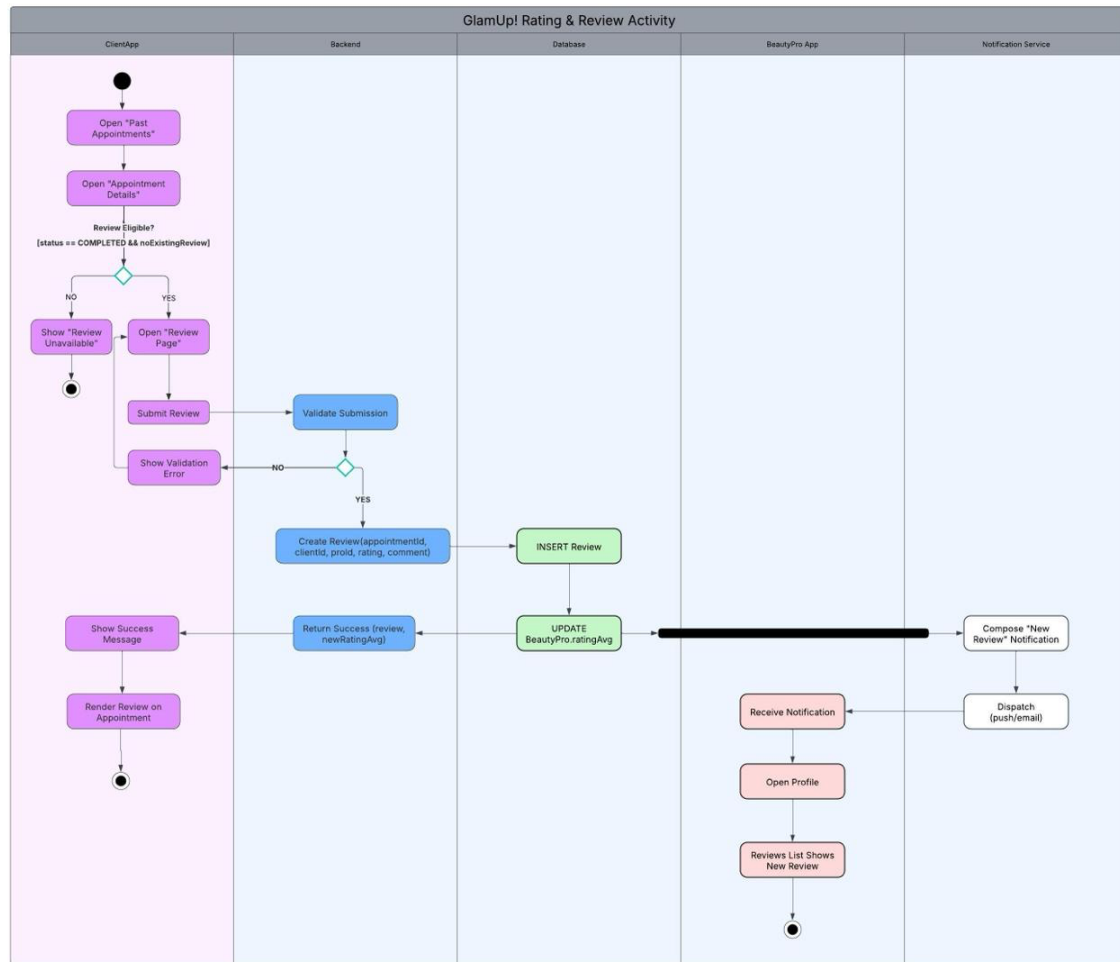
Beauty Professional Activity Diagram



Reporting User Activity Diagram:**Reporting User - Activity Diagram (GlamUp!)**

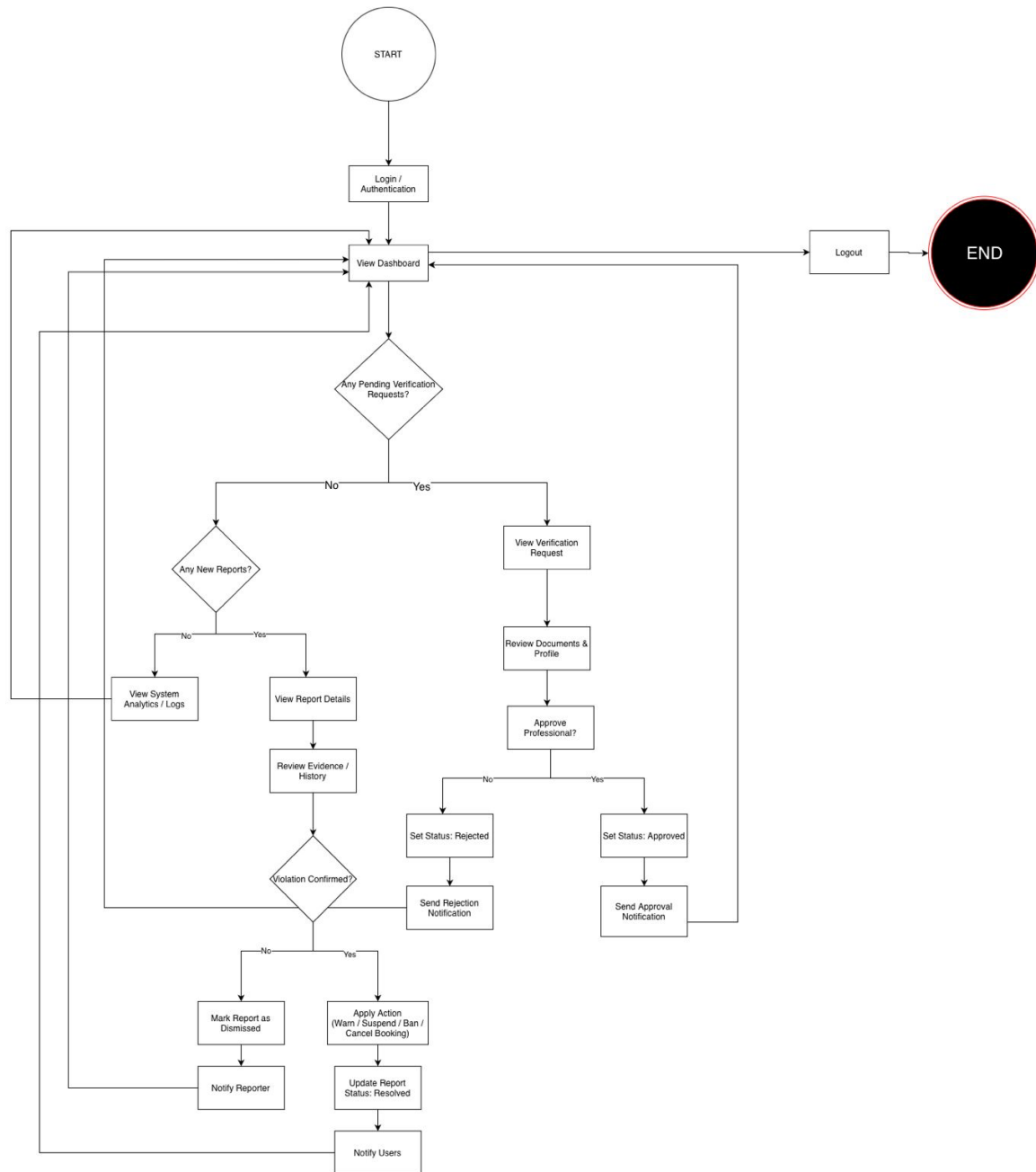
```
graph TD
    subgraph Client
        C1(( )) --> C1_1[Open GlamUp]
        C1_1 --> C1_2[Enter login credentials]
        C1_2 --> C1_3[Open bookings / messages]
        C1_3 --> C1_4[Select booking / conversation]
        C1_4 --> C1_5[View chat history]
        C1_5 --> C1_6[Type message]
        C1_6 --> C1_7[Send message]
        C1_7 --> C1_8[Read reply]
        C1_8 --> C1_9{Client sends new message?}
        C1_9 -- yes --> C1_10[Type new message]
        C1_9 -- no --> C1_11{ }
        C1_10 --> C1_12[Send new message]
        C1_12 --> C1_11
        C1_11 --> C1_13[Close chat / Logout]
    end

    subgraph GlamUp_System [GlamUp System]
        G1[Validate client credentials] --> G2{Client credentials valid?}
        G2 -- yes --> G3[Create client session]
        G2 -- no --> G4[Show error message]
        G4 --> G4_1(( ))
        G3 --> G5[Load client conversations]
        G5 --> G6[Return conversation list]
        G6 --> G7[Validate sender & booking]
        G7 --> G8[Store message in DB Messages]
        G8 --> G9[Mark as unread for Professional]
        G9 --> G10[Trigger notification to Professional]
        G10 --> G11[Validate pro credentials]
        G11 --> G12{Pro credentials valid?}
        G12 -- yes --> G13[Create pro session]
        G12 -- no --> G14[Show error message]
        G14 --> G14_1(( ))
        G13 --> G15[Load conversation history]
        G15 --> G16[Return messages]
        G16 --> G17[Validate pro & booking]
        G17 --> G18[Store reply in DB Messages]
        G18 --> G19[Notify Client]
        G19 --> G20[Validate sender & booking]
        G20 --> G21[Store message in DB Messages]
        G21 --> G22[Notify Professional]
        G22 --> G23[Validate sender & booking]
        G23 --> G24[Store message in DB Messages]
        G24 --> G25[Notify Professional]
        G25 --> G26[Validate sender & booking]
        G26 --> G27[Store message in DB Messages]
        G27 --> G28[Notify Professional]
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        G128 --> G129[Store message in DB Messages]
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        G141 --> G142[Notify Professional]
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        G190 --> G191[Validate sender & booking]
        G191 --> G192[Store message in DB Messages]
        G192 --> G193[Notify Professional]
        G193 --> G194[Validate sender & booking]
        G194 --> G195[Store message in DB Messages]
        G195
```

Ratings & Reviews Activity Diagram:

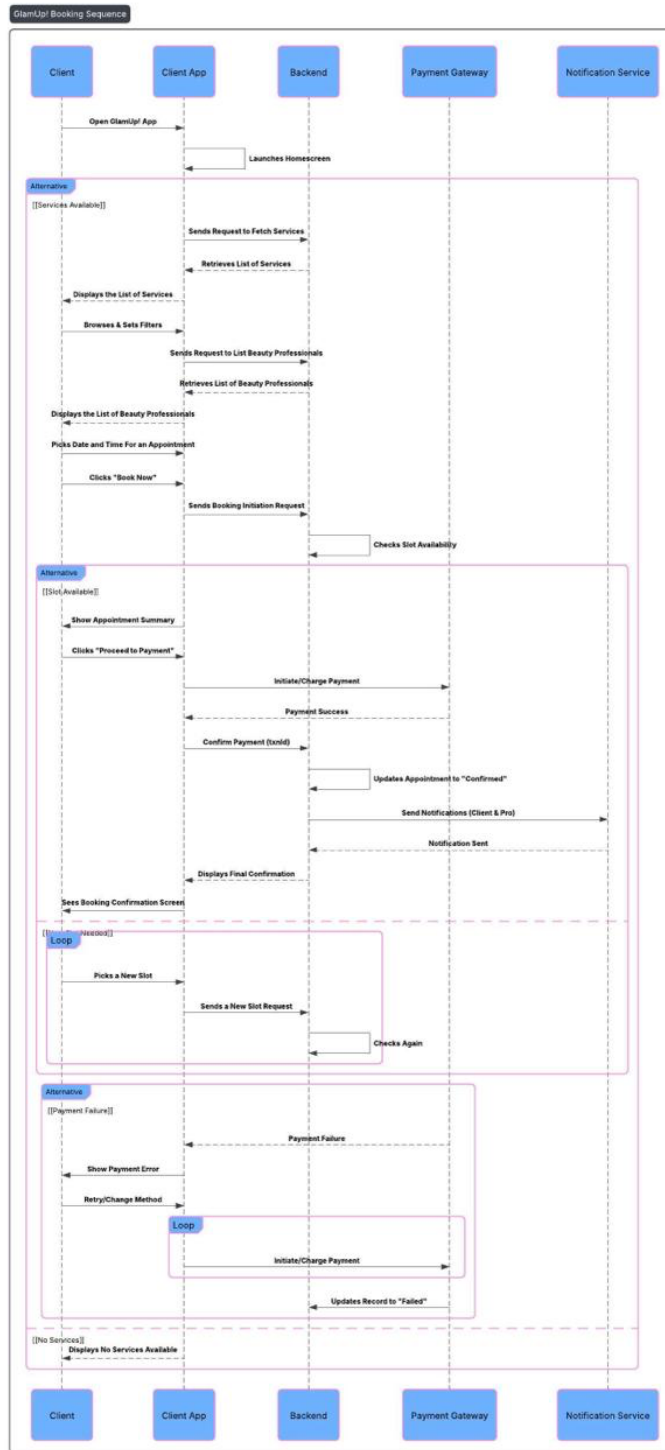
Admin Activity Diagram:

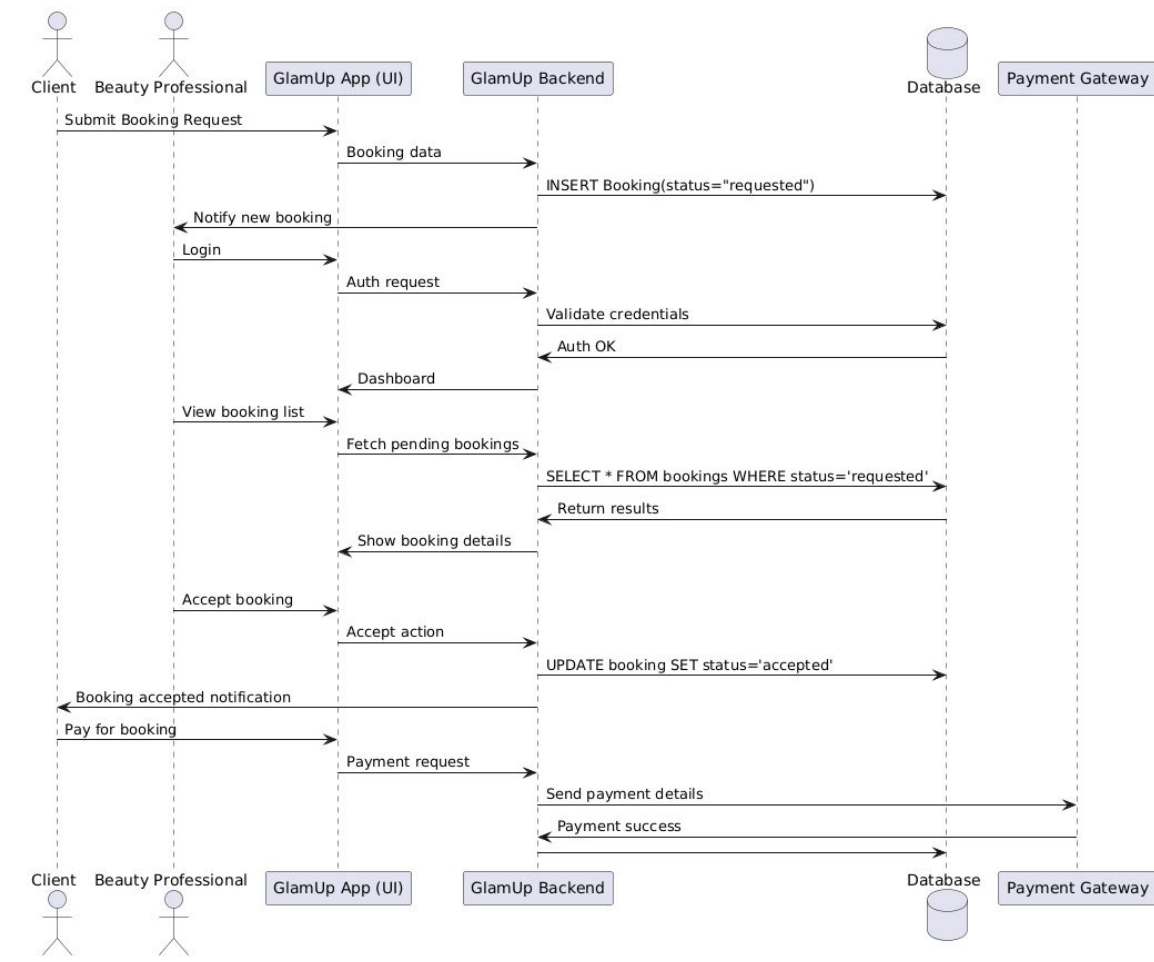
Admin Activity Diagram

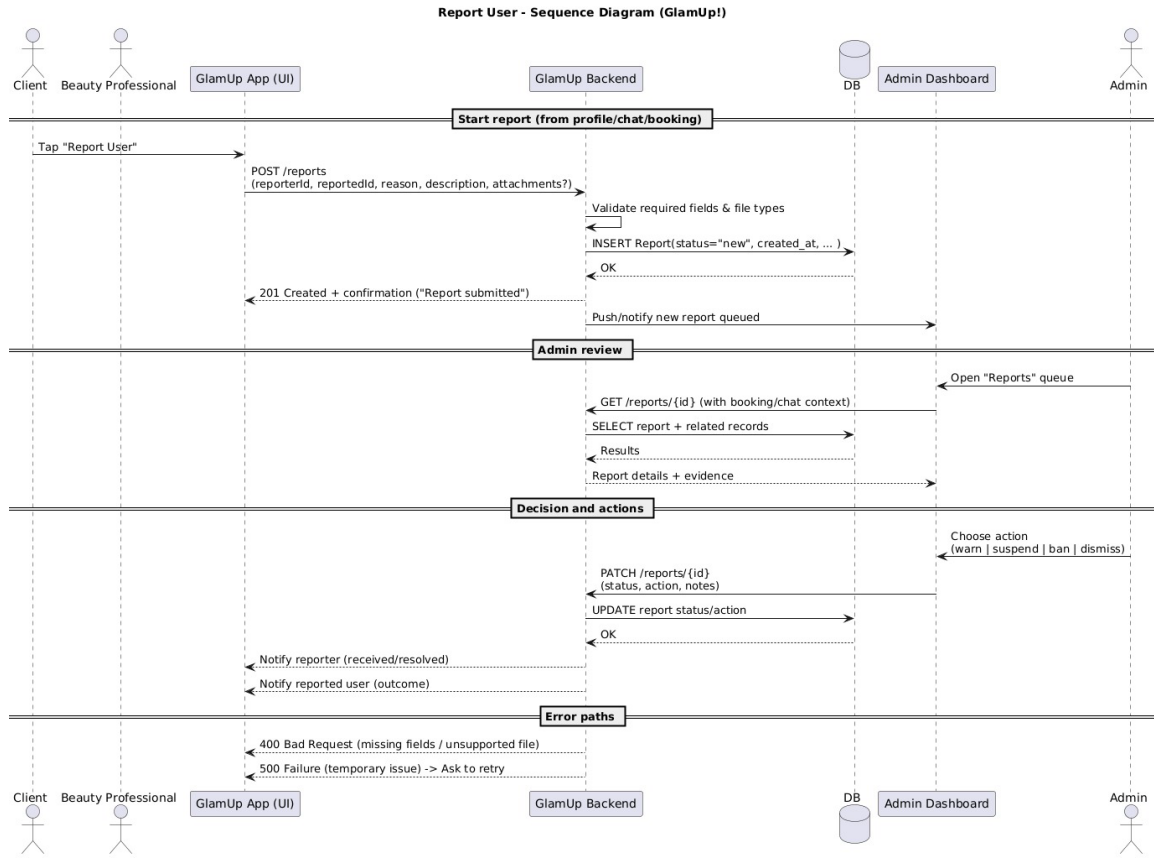


3.3.3 Sequence Diagrams

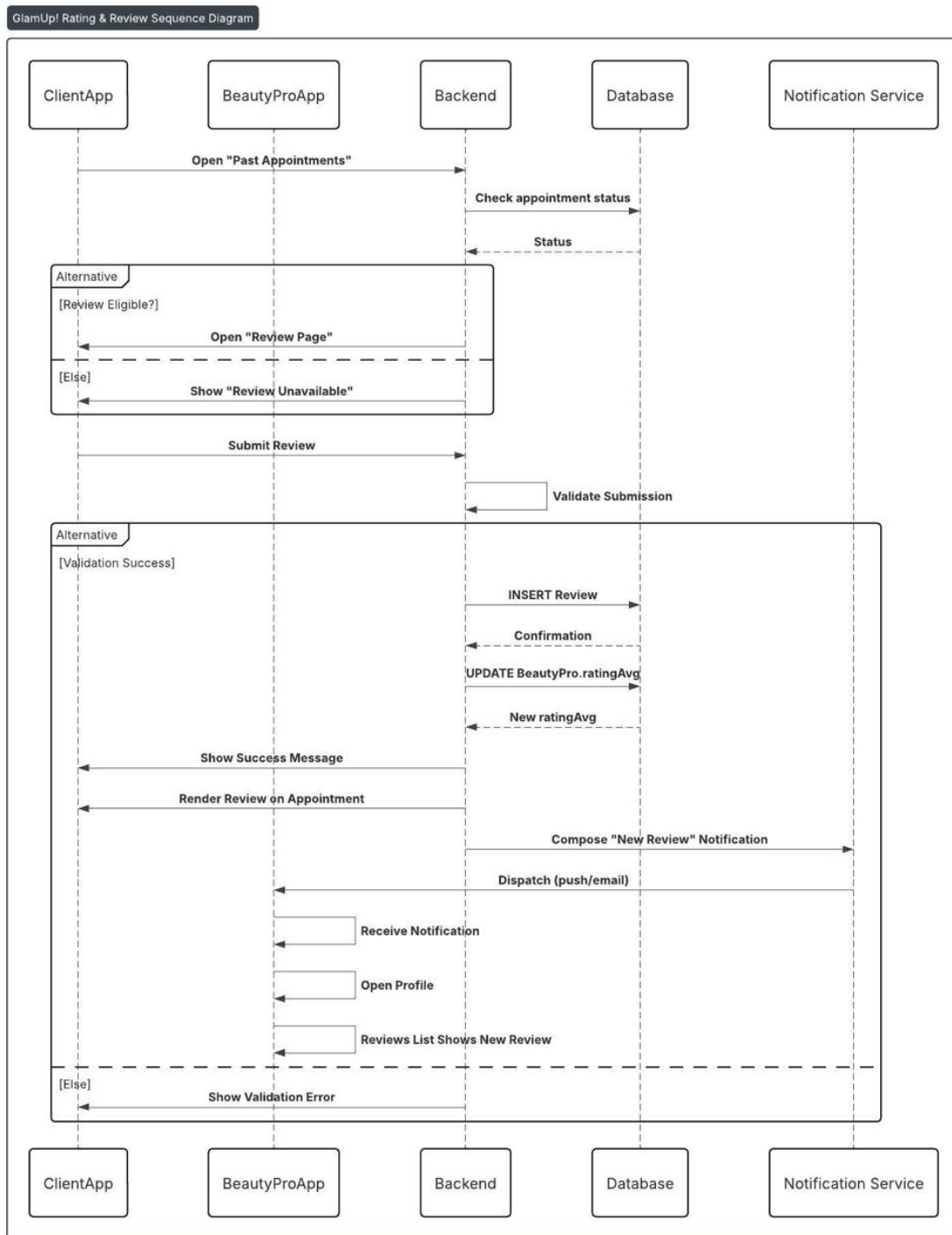
Client Booking Sequence Diagram

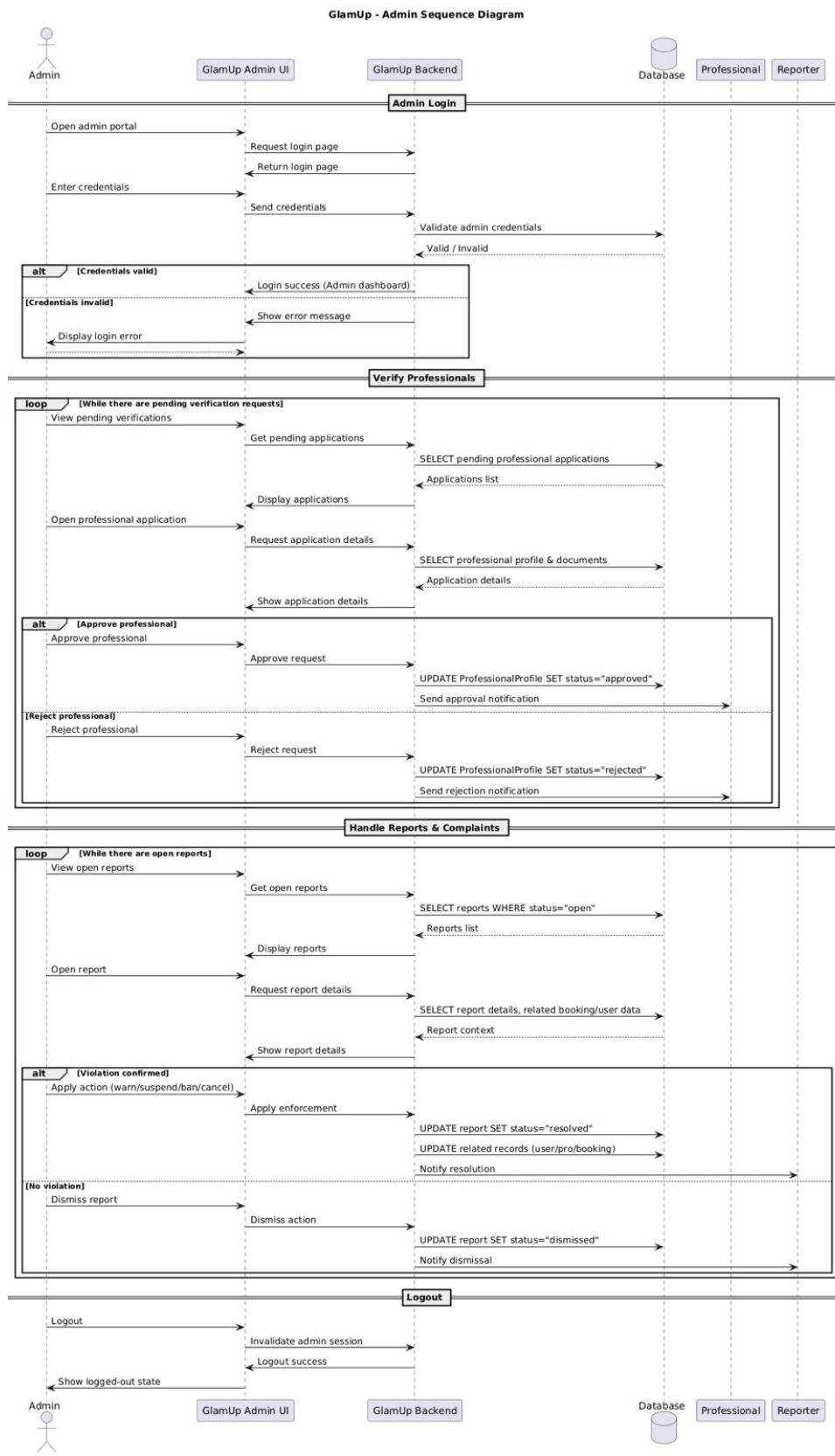


Beauty Professional Sequence Diagram:

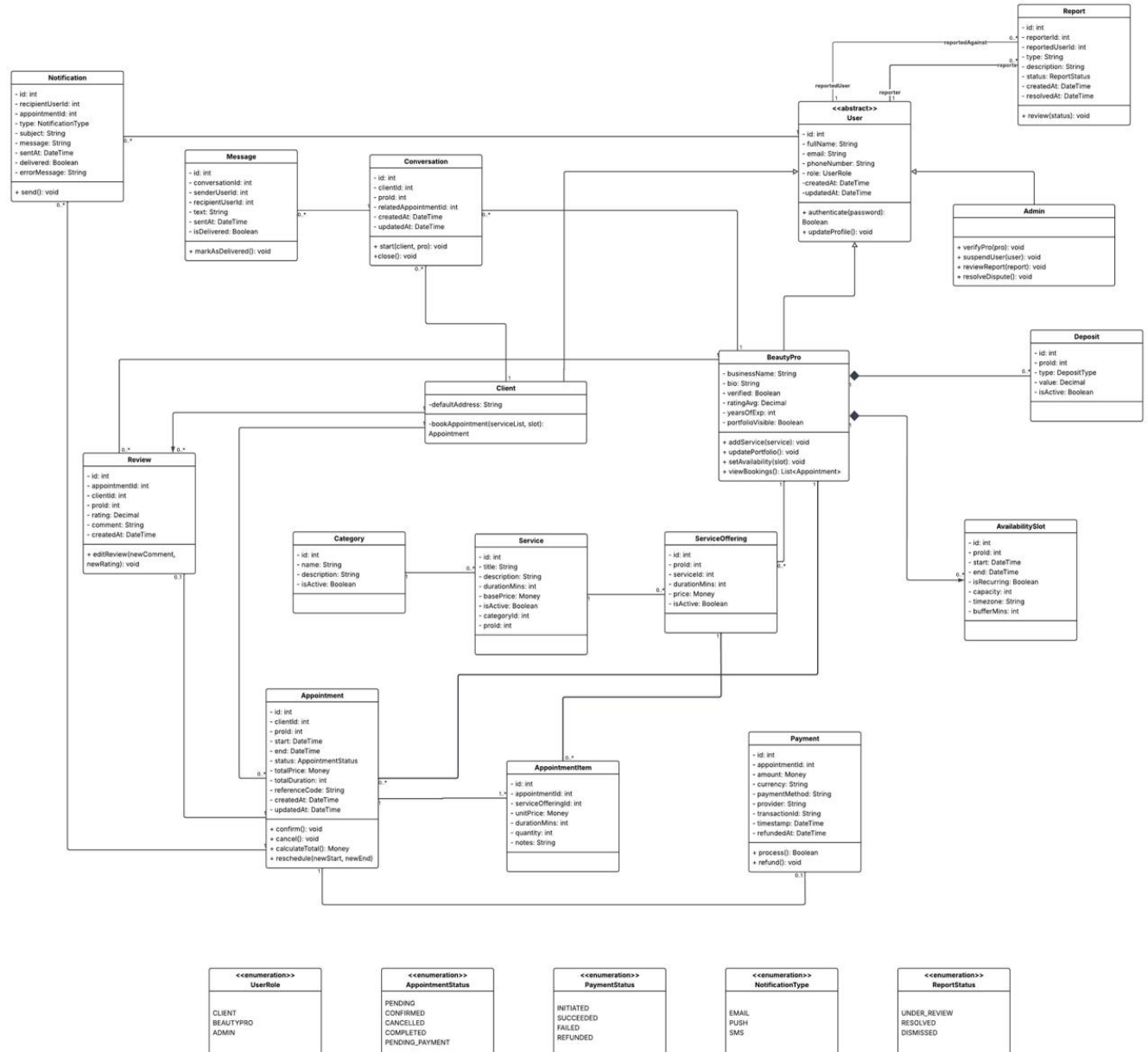
Report User Sequence Diagram:

Private Messaging Sequence Diagram:

Ratings & Review Sequence Diagram:

Admin Sequence Diagram:

3.3.4 UML Class Diagram

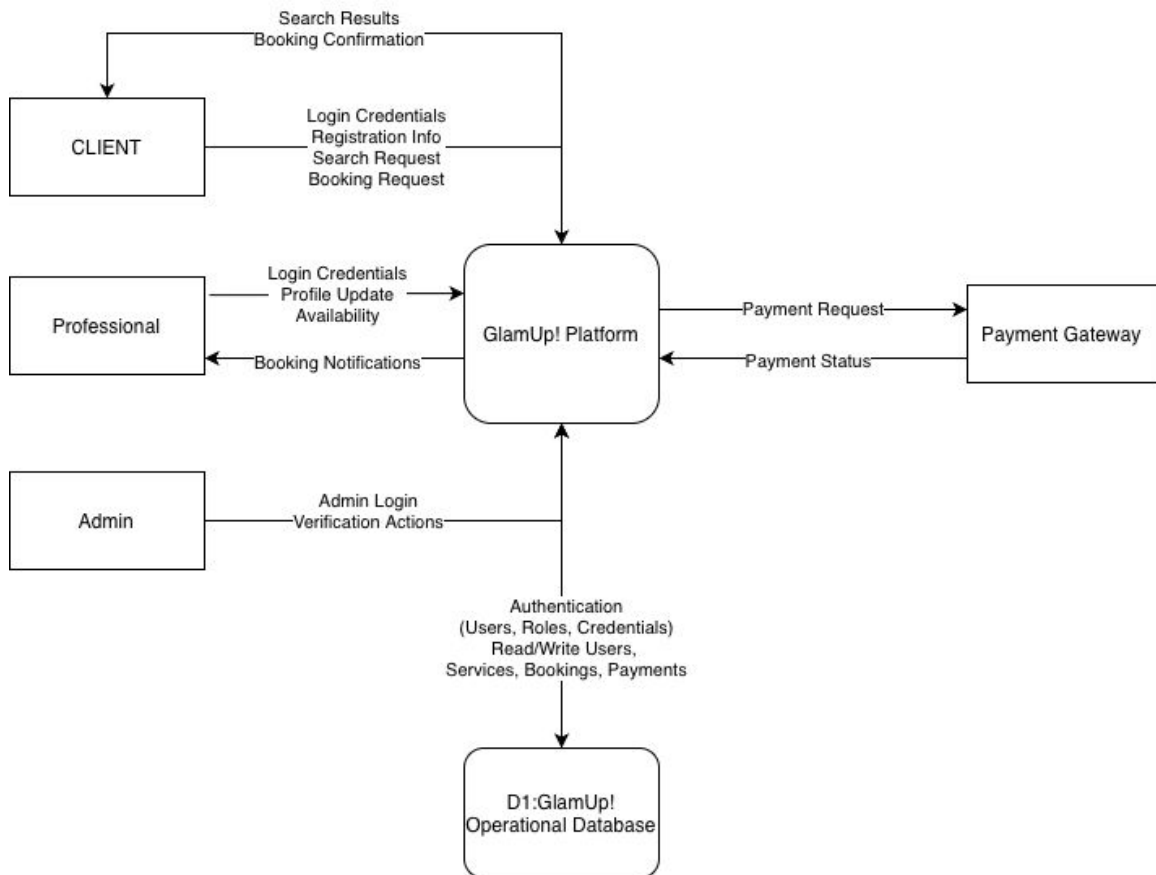


3.4 Process Modelling

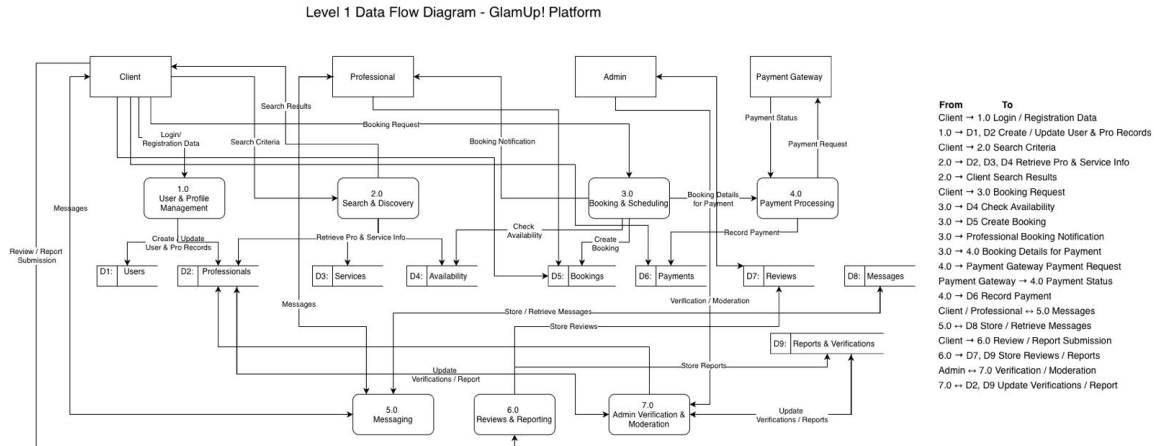
3.4.1 Level 0 Data Flow Diagram

The Level 0 Data Flow Diagram illustrates the high-level interactions between external entities and the GlamUp Platform. Clients, Professionals, and Administrators each authenticate with login credentials before sending or receiving data. The central GlamUp Platform manages requests, communicates with the Payment Gateway for secure transactions, and reads/writes information to the GlamUp Operational Database. This diagram summarizes the main system boundaries and data exchanges for booking, payments, verification, and user management.

Level 0 Data Flow Diagram – GlamUp Platform



3.4.2 Level 1 Data Flow Diagram



4.0 Non-Functional Requirements

Performance

- 95% of search results must return in < 1.0s for datasets up to 50k professionals.
- Booking create/update operations must complete in < 2.0s end-to-end (incl. payment auth).

Reliability & Availability

- Target uptime $\geq 99.5\%$ per month.
- Any failed payment or booking write must be atomically rolled back and retried up to 3 times with exponential backoff.

Security & Privacy

- All data in transit via HTTPS/TLS 1.2+; PII and payment tokens encrypted at rest (AES-256).
- RBAC: client, professional, and admin roles with least-privilege access.
- No card data stored; use PCI-compliant gateway tokens only.
- Audit log for all admin actions; retention 12 months.

Maintainability

- Service modules (Auth, Search, Booking, Payments, Reviews) isolated; APIs documented.
- Unit test coverage $\geq 80\%$; CI on every PR.

Scalability

- Read-heavy features (search, profiles) backed by indexes + cache (e.g., Redis).
- Must scale to 100 concurrent bookings/min without degradation.

Portability

- Containerized deployment (Docker); cloud-agnostic infra (PostgreSQL/Mongo-compatible).

5.0 Logical Database Requirements

5.1 Core Entities (logical model)

- User(user_id PK, role {client|pro|admin}, name, email unique, phone, hashed_password, created_at, status)
- ProfessionalProfile(pro_id PK/FK→User, bio, services_offered[], base_location (lat, lng), portfolio_urls[], rating_avg, rating_count, verification_status)
- Service(service_id PK, pro_id FK→ProfessionalProfile, name, category {nails|lashes|makeup|...}, duration_min, price_cents, is_active)
- Availability(availability_id PK, pro_id FK, weekday, start_time, end_time, timezone)
- Booking(booking_id PK, client_id FK→User, pro_id FK→ProfessionalProfile, service_id FK→Service, start_ts, end_ts, status {pending|confirmed|completed|cancelled|refunded}, price_cents_snapshot)
- Payment(payment_id PK, booking_id FK→Booking, gateway {Stripe|PayPal}, token, amount_cents, currency, status {authorized|captured|failed|refunded}, created_at)
- Review(review_id PK, booking_id FK→Booking, client_id FK→User, pro_id FK→ProfessionalProfile, stars 1–5, comment, created_at, is_flagged)
- Location(location_id PK, address, city, province/state, country, postal_code, lat, lng) — optional if storing structured addresses.

5.2 Data Formats

- Monetary values stored as integer cents; UTC timestamps (ISO-8601 at the API).
- Images/portfolio stored as URLs to object storage; not in DB blobs.

5.3 Integrity Rules

- A Booking must reference an existing Client, Professional, and Service.
- Review allowed only when Booking.status = completed and author = booking.client_id.
- Payment requires unique (booking_id, status=captured) to prevent double-charge.
- ProfessionalProfile requires verification_status before services become bookable.

2.2 5.4 Indexing & Query Needs

- Index on ProfessionalProfile(base_location) with geospatial support for “near me” search.
- Composite indexes:
 - Service(pro_id, category, is_active)
 - Booking(pro_id, start_ts) and Booking(client_id, start_ts)
 - Review(pro_id, created_at)
- Full-text index on services_offered/name for keyword search.

5.5 Storage & Retention

- Bookings/Payments: retain 7 years (financial/audit).
- Reviews: retain indefinitely unless flagged/removed per policy.
- Soft delete users/profiles; hard delete PII on verified GDPR/CCPA requests.

5.6 Consistency

- Use transactional writes for booking + payment authorization (saga pattern acceptable across services).
- Idempotency keys on booking creation and payment capture.

5.7 Backup & Recovery

- Automated daily backups; point-in-time recovery ≤ 15 min RPO, ≤ 1 hr RTO.

6.0 Other Requirements

- Accessibility: WCAG 2.1 AA for client-facing flows (color contrast, keyboard nav, alt text).
- Localization: Support EN initially; structure copy for easy addition of FR.
- Notifications: Email + push for booking confirmations, reminders (T-24h, T-2h), cancellations, and payout notices.
- Legal/Compliance: Terms of Service, Privacy Policy, consent for marketing emails; content moderation for reviews (auto-flag profanity/abuse).
- Analytics: Non-PII events for search funnels, conversion, and churn; configurable via feature flags.

7.0 Approval

The signatures below indicate their approval of the contents of this document.

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